Conditional Hilbert Lifts of ℓ^1 Sector Weights and Born Normalization Redundancy

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Abstract

This paper constructs a conditional analysis of Born normalization as a redundant Hilbert-space encoding of already-given nonnegative ℓ^1 sector weights. The paper proves direct probability normalization from additive weights, shows that the square-root Hilbert lift reproduces the same law, and quarantines Hilbert dynamics, phase averaging, measurement descent, and many-worlds branch-localization claims as conditional or conjectural.

1 Central Thesis

The central thesis of this paper resolves to the following redundancy bound:

> The paper does not derive quantum mechanics from ℓ^1; it demonstrates that once non-negative gauge-reduced sector weights are established via Disjoint Observer Consistency, the Born normalization formula is reproduced by an optional Hilbert lift (governed by Conditional Minimality) without adding new fundamental ontic content.

This separates the results into a cleanly verifiable rigorous core (Layer A), supported by two distinct interpretive layers.

1.1 Classification of Results

Layer	Status	Content
A: Theorem & Redundancy	Rigorous	Direct probability law from weights, Hilbert-lift redu
B: Structural Interpretation	Interpretive	Measurement as fixed-point descent of the topologica
C: Graph-Support Sketch	Conditional Sketch	Argument predicting Many-Worlds branch blockage.

LAYER A: WEIGHTS, LIFTS, AND BORN RECONSTRUCTION

2 The ℓ^1 Sector Weights and Direct Measurement

The ℓ^1 coboundary framework yields finite residual sectors Γ carrying weights w_γ. These weights are mathematically non-negative, additive on disjoint sectors, and gauge-reduced.

Theorem 2.1 (Direct Normalization). Let Γ be a finite set of residual sectors with weights $w_\gamma \geq 0$, not all zero. The quantity

$$P_{\ell^1}(\gamma) := \frac{w_\gamma}{\sum_{\eta\in\Gamma} w_\eta}$$

defines a direct normalized probability distribution on Γ.

3 The Uniqueness of ℓ^1 Additive Bookkeeping

Proposition 3.1 (Nonlinear Aggregation for $p > 1$). ℓ^1 is the unique ℓ^p norm with exact additive bookkeeping on disjoint positive sectors, while $p > 1$ introduces nonlinear aggregation.

If disjoint sectors x and y are non-interacting, their weights sum linearly: $\|x \oplus y\|_1 = \|x\|_1 + \|y\|_1$. For any $p > 1$, we predictably observe strict subadditivity: $\|x \oplus y\|_p < \|x\|_p + \|y\|_p$.

4 Categorical Formalization: Lift and Descent Maps

If we voluntarily embed the ℓ^1 weights into a complex representation computing an amplitude lift $\left(c_\gamma = \sqrt{w_\gamma} e^{i\theta_\gamma}\right)$, we formalize this mapping by establishing topological boundaries.

4.1 Descent Equivalence and the Quotient Factorization

We define the algebraic tracking of phase-blindness via an equivalence relation $\sim$:

$$c \sim c' \iff |c_\gamma|^2 = |c'_\gamma|^2 \quad \forall \gamma \in \Gamma$$

Let D_Γ be the pointwise descent evaluation $D_\Gamma(c) = |c|^2$. Because D_Γ is identically constant on $\sim$-equivalence classes, it formally factors through the quotient:

$$D_\Gamma = \bar{D}_\Gamma \circ q_\Gamma$$

where $\bar{D}_\Gamma$ isolates exactly the ontic modulus data.

4.2 The Row-Sparsity Lemma and Compatible Lifts

Lemma 4.1: Strict Row-Sparsity of Descent-Compatible Maps. Let U be a linear map structurally satisfying descent compatibility. Each output coordinate of U can mathematically depend upon at most one input coordinate.

4.3 The Born Re-encoding and Modulus Minimality

The descent functor $\mathsf{D} : \mathbf{Lift}_{\text{comp}} \to \mathbf{Wt}$ maps exactly via $\mathsf{D}(\Gamma, c) := \left(\Gamma, |c|^2\right)$.

Theorem 4.2 (Hilbert-Lift Redundancy). Let Γ carry nonnegative sector weights $w_\gamma \geq 0$, and let $c_\gamma = \sqrt{w_\gamma} e^{i\theta_\gamma}$. The standard Born rule executed on the lifted state identically recovers the direct ℓ^1 measure:

$$P_{\text{Born}}(\gamma) := \frac{|c_\gamma|^2}{\sum_\eta |c_\eta|^2} = \frac{w_\gamma}{\sum_\eta w_\eta} \equiv P_{\ell^1}(\gamma).$$

5 Phase-Averaged Quadratic Observables

Theorem 5.1 (Phase-Averaged Shadow of the Lifted Encoding). Let $|\psi(\theta)\rangle = \sum_{\gamma\in\Gamma} \sqrt{w_\gamma} e^{i\theta_\gamma}|\gamma\rangle$, and let $|u\rangle = \sum_\gamma u_\gamma|\gamma\rangle$. Then:

$$\mathbb{E}_{\theta\sim\mathrm{Haar}(U(1)^{|\Gamma|})}\left[|\langle u, \psi(\theta)\rangle|^2\right] = \sum_\gamma |u_\gamma|^2 w_\gamma.$$

LAYER B: INFERENTIAL STRUCTURE AND NATIVE DECOMPOSITION

6 The Mathematical Dictionary: ℓ^2 Hodge Shadow vs. ℓ^1 Native Split

- ℓ^2 **(Hodge):** $\delta = Bf_2^* + h_2$ (Least-squares orthogonal projection)
- ℓ^1 **(Native):** $\delta = Bf_1^* + r_1$ (Quotient-minimizing primal split)

The structural analogue to harmonic continuous structure requires substituting the **dual witness** p^*, where $B^\top p^* = 0$ and $\langle p^*, \delta\rangle = \Phi([\delta])$.

7 Measurement Model as Fixed-Point Descent

Interpretive Proposition (Layer B): If structural observation aligns with repeated exact repair/projection, then measurement may be modeled as fixed-point attainment of T rather than a catastrophic non-local wave collapse.

7.1 Conditional Quadratic Lift

Assume: (1) finite sector weights $w_\gamma \geq 0$ with additive normalization; (2) a continuous complex linear representation $\psi \in \mathbb{C}^\Gamma$; (3) a phase-blind positive descent map $F(\psi)$; (4) orthogonal additivity; (5) reversible lifted dynamics; and (6) interference represented by bilinear cross terms. Under these assumptions, the scalar descent is quadratic: $F(\psi)_\gamma \propto |\psi_\gamma|^2$.

7.2 Unitary Evolution as Obstruction-Preserving Automorphism

Unitaries are the mass-preserving automorphisms of the ℓ^2 envelope of ℓ^1 obstruction. U does not repair h_1; it redistributes obstruction mass reversibly in the lifted phase space.

7.3 Commutator as Lift-Project Path Mismatch

The quantity $K(A, B; \psi) := \||\psi_{AB}|^2 - |\psi_{BA}|^2\|_1$ measures order-dependence visible after $|\cdot|^2$ descent to ℓ^1.

LAYER C: A GRAPH-SUPPORT OBSTRUCTION SKETCH

8 Geometric Definitions for Candidate Local Sectors

Conjecture 8.1 (Functorial Localization Failure for Non-Duplicable Support). Let G be a dependency graph carrying a graph-supported irreducible class with globally non-duplicable minimal support. No admissible branch localization (G_A, G_B) exists.

9 Conclusion

Layer A proves a narrow redundancy result: once nonnegative ℓ^1 sector weights are given, Born normalization adds no probability information. Layer B is conditional on a Hilbert lift. Layer C is a graph-localization conjecture.

A Conditional Uniqueness of the Quadratic Lift

Any scalar valuation map $F(c)$ satisfying orthogonal additivity, continuous symmetric invariance, and scalar normalization over a unitary space is mathematically confined to the direct quadratic form: $F(c) \propto |c|^2$.

Changelog

v007 Audit Note

- ℓ^1 weights alone do not force Hilbert space.
- Noncommuting lifted operators are not equivalent to observable order mismatch.
- Decoherence removes phase-sensitive interference under a fixed pointer-basis restriction.